

Who Will AI Displace?

Structural Barriers, Demand Traps, and the Limits of Early Evidence on AI-Driven Labor Displacement

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Executive Summary

Two independent research teams — Massenkoff and McCrory at Anthropic (2026) and Hemenway Falk and Tsoukalas at Penn/BU (2026), converge on a counter-intuitive finding: **the primary barrier to AI-driven labor displacement is not technical capability but structural economic contradictions.**

- *The Empirical Signal:* The Anthropic paper provides early evidence of a shift hitting highly educated, well-paid workers disproportionately.
- *The Theoretical Proof:* The Penn/BU paper proves formally that competitive firms will rationally over-automate into a Pareto-inferior equilibrium that only a Pigouvian tax can correct.

Together they identify a policy problem more troubling than job loss itself: the evidence required to politically justify intervention may arrive only after the trap has already closed. Both papers, however, share some structural blind spots. The demand trap may be deferred through displaced workers' savings capacity, and aggregate unemployment conceals occupational downgrading as the primary adjustment channel.

Core Idea of the Material

The conventional discourse on AI focuses on capability, particularly which tasks models can perform and how close they are to achieving human level or general intelligence. When AI falls short, it is often explained through technical limitations such as hallucinations or delays in real world implementation. However, recent academic work suggests that a deeper mechanism is at play. The key constraint is not what AI can do, but the economic and structural incentives that determine how, when, and to what extent firms choose to deploy it, and what this implies for aggregate demand.

Massenkoff and McCrory (2026) introduce a new measure of AI displacement risk, *Observed Exposure*, which combines O*NET task data with actual Claude usage patterns from Anthropic's Economic Index and theoretical feasibility scores from [Eloundou et al. \(2023\)](#). The key innovation is the gap it reveals. AI's theoretical exposure (as measured by [Eloundou et al.](#)

(2023)) covers nearly 80% of the US workforce; while observed AI deployment is constrained by model limitations, legal requirements, and organizational inertia. Most notably, it covers a far smaller share. The empirical findings so far are ambiguous. Aggregate unemployment among high exposure workers has not risen significantly. At the same time, occupations with higher observed exposure are projected to experience weaker growth over the next decade. A more immediate signal appears among younger workers in exposed occupations, who show a statistically significant 14 percentage point decline in monthly job finding rates relative to 2022. The authors interpret this as early evidence of a shift in labor demand, but caution that it is too soon to draw firm conclusions.

Hemenway Falk and Tsoukalas (2026) work at a different level of abstraction but arrive at a structurally similar conclusion. They build a task-based model in the classical sense of [Acemoglu and Restrepo \(2020\)](#) and show that firms have strong incentives to over-automate, leading to worker displacement well beyond the social optimum. The mechanism operates through a demand externality. As firms adopt AI and replace parts of their workforce, the purchasing power of displaced workers declines. While it may seem automation lowers production costs, it simultaneously erodes the demand on which firms depend. Individual firms capture the cost savings from automation but absorb only a small share of the resulting decline in aggregate demand. This misalignment of incentives actually drives firms to automate beyond the socially optimal level.

The resulting Nash equilibrium is Pareto dominated. Both workers and firm owners are worse off than under the cooperative outcome. Most notably, the authors prove that standard redistributive policies such as UBI, capital taxes, profit sharing, and Coasian bargaining fail to address this underlying incentive misalignment. Only a Pigouvian automation tax can restore the cooperative optimum.

***The counter-intuitive convergence:** The barrier to AI replacing workers is not that AI cannot do the work; it is that the economic structure governing deployment guarantees firms will over-automate even if they can collectively foresee the harm — and that standard policy instruments are powerless to stop them.*

Extension with Additional Sources

To situate the structural claims of both papers, it helps to trace how the economics of technological displacement has evolved.

The foundational task-based framework of [Autor et al. \(2003\)](#) established that computerization substitutes for routine cognitive and manual tasks while complementing non-routine analytical ones, a pattern they show accounts for a significant college wage premium for decades. Generative AI disrupts precisely this boundary, targeting non-routine cognitive work such as drafting, synthesis, structured reasoning, code generation — the category [Autor et al. \(2003\)](#) had identified as structurally safe. This disruption explains why [Acemoglu and Restrepo \(2020\)](#)'s seminal framework on industrial robots cannot serve as a direct template. Their robots-and-jobs framework was built for a technology that displaced low-skill workers, but complemented high-skill labour. [Hemenway Falk and Tsoukalas \(2026\)](#) extend it by

adding a new macroeconomic demand channel that [Acemoglu and Restrepo \(2020\)](#) did not take into account. The omission made sense for industrial robots, as displaced factory workers were not the primary buyers of industrial goods; however, the high-wage knowledge workers threatened by AI are the primary consumers driving the service economy.

This historical shift also resolves the Anthropic paper's most puzzling finding: the massive gap between AI's theoretical capacity and its actual deployment. While [Eloundou et al. \(2023\)](#) estimate that 80% of US workers face exposure, Anthropic's study reveals that observed usage covers only a fraction of this ceiling. Rather than offering reassurance, this gap is a classic signature of general-purpose technology adoption. Like electricity or the internet before, generative AI requires profound organizational restructuring before productivity gains materialize at scale. Viewed through this lens, the low displacement rates in the Anthropic data reflect the friction of early integration, not a stable equilibrium. And [Acemoglu \(2024\)](#)'s modest estimate of a 0.71% productivity gain over the next decade is best understood not as a ceiling on AI's eventual reach, but as a near-term projection for a technology still in its organizational infancy.

At the firm level, the clearest evidence of this dynamic comes from [Brynjolfsson et al. \(2023\)](#). Studying 5,179 customer-support agents, they document a 14% average productivity increase, which leaps to 34% for the least experienced workers. The mechanism is knowledge diffusion: AI codifies the best practices of top performers and makes them available to everyone. While this is often seen as a positive form of support, this dynamic contains a structural threat that aligns well with the trap [Hemenway Falk and Tsoukalas \(2026\)](#) formalize. Once this knowledge is built into the system, firms have less reason to hire and train junior workers. This shows that augmentation and displacement are not mere alternatives, but sequential phases of the process.

This pattern also appears in broader institutional forecasts. For example, while the [IMF \(2024\)](#) warns of 60% AI exposure in advanced economies, [Goldman Sachs \(2023\)](#) encapsulates the macroeconomic counterargument perfectly. They estimate that despite a staggering 300 million jobs globally are exposed to AI, they project a 7% uplift in global GDP, concluding that long-term job creation will ultimately outpace immediate displacement and the [World Economic Forum \(2025\)](#) project massive net job creation over the next decade. Yet, the WEF simultaneously concedes that 40% of employers are already planning workforce reductions. This contradiction reveals a severe structural asymmetry — the macroeconomic benefits of AI are projected over a decade, while the microeconomic incentives to cut headcount are immediate and already in place.

Key Tensions and Gaps

1. The Demand Externality Requires a Consumer Profile the Data Does Not Supply.

The demand trap in [Hemenway Falk and Tsoukalas \(2026\)](#) rests on a precise empirical assumption: that displacing workers reduces household expenditure enough to make automation collectively self-defeating. The mechanism requires displaced workers to behave as high marginal propensity to consume agents. [Massenkoff and McCrory \(2026\)](#)'s exposure data challenges this directly. The workers most at risk earn 47 percent more

per hour than unexposed workers and are disproportionately graduate-educated, precisely the household profile that saves more and spends proportionally less out of each additional dollar of income. High-skilled displaced workers can draw on financial assets and severance to cushion their consumption far longer than the demand model anticipates. The trap may be real but substantially deferred, arriving through gradual wealth depletion rather than an immediate income shock, and visible in aggregate data only after the structural damage has compounded. This has a direct implication for monitoring. The leading indicator for the trap is not a consumption collapse but a slow erosion in household balance sheets, a signal that existing macroeconomic infrastructure was not designed to track.

2. Aggregate Unemployment Is the Wrong Diagnostic for High-Skill Displacement.

The Anthropic paper treats the absence of significant aggregate unemployment among highly exposed workers as a preliminary reassuring signal. This reading fundamentally misunderstands how high-skill labor markets absorb displacement pressure. Highly educated, well-compensated workers who lose primary roles rarely transition directly into sustained unemployment. Instead, they could experience occupational downgrading, in which they accept positions below their credential level, moving into roles previously held by workers one tier down the skill distribution. This creates a cascading effect across the workforce. Displaced senior workers take mid-level positions, which then pushes mid-level workers into entry-level roles. This process still triggers the income loss described by the [Hemenway Falk and Tsoukalas \(2026\)](#) demand model, but it registers as wage stagnation rather than a spike in official jobless claims. As a result, this displacement remains entirely invisible to the aggregate indicators the Anthropic paper relies on to assess early impact. The true warning signal, which neither paper actually measures, is *underemployment* — the growing share of educated workers forced into roles below their credential level.

3. The Prisoner's Dilemma Logic Does Not Hold in the Concentrated Markets Where AI Adoption Is Empirically Concentrated.

The over-automation result in [Hemenway Falk and Tsoukalas \(2026\)](#) assumes a perfectly competitive market where firms face a prisoner's dilemma. They either automate to survive price competition, or be undercut by rivals. This mechanism requires firms to lack pricing power, making automation a survival imperative rather than a strategic choice. However, empirical literature shows this assumption breaks exactly where it matters most. AI deployment is heavily concentrated among dominant "superstar" firms in oligopolistic industries ([Autor et al., 2020](#); [IMF, 2024](#)). A firm insulated by market power does not face the competitive discipline that activates the prisoner's dilemma: it can pace automation to maximize long-run profitability and absorb productivity gains as expanded margins rather than rapidly deploying AI to survive price competition. [Massenkoff and McCrory \(2026\)](#) completely obscure this dynamic by aggregating their exposure data without adjusting for firm size or market power. Consequently, the forced over-automation that drives the Penn/BU model may be completely absent among the very firms leading AI deployment. A universal Pigouvian tax based on these models could result in a severe policy mismatch by heavily taxing dominant firms that are already

pacing their rollout while failing to address the competitive fringe where the prisoner's dilemma actually exists.

4. Neither Paper Theorizes the Path From Augmentation to Displacement, Which Is Where Policy Must Intervene.

The two papers describe different temporal states without connecting them. [Massenkoff and McCrory \(2026\)](#) document current conditions, where observed AI deployment falls far short of its theoretical ceiling and aggregate unemployment has not responded, and interpret this as a leading signal. [Hemenway Falk and Tsoukalas \(2026\)](#) characterize a future equilibrium in which automation has surpassed the critical threshold. Neither paper theorizes the transition between these states. General-purpose technology theory, from [Helpman and Trajtenberg \(1998\)](#)'s foundational framework to [Brynjolfsson et al. \(2021\)](#)'s productivity J-curve, consistently finds that GPT adoption is non-linear: early deployment generates reorganization costs and apparent stagnation before complementary innovations in workflows and human capital allow gains to diffuse. If AI follows this pattern, the current data reflects the bottom of the J-curve, the phase where firms reorganize and slow entry-level hiring before new tasks emerge. Deploying a Pigouvian tax at the bottom of the J-curve would suppress the complementary investment that eventually resolves the displacement phase, while withholding it until the equilibrium is directly observable may mean the trap has already closed.

Four structural gaps: First, the demand trap is likely deferred rather than averted, as high-earning displaced workers can temporarily sustain consumption by drawing on savings and assets while structural damage compounds. Second, aggregate unemployment may be the wrong diagnostic, as the true adjustment occurs through occupational downgrading, which remains invisible to the indicators both papers rely on. Third, the over-automation mechanism may be overstated, as prisoner's dilemma logic need not bind in the oligopolistic settings where AI adoption is concentrated. Finally, these gaps point to a deeper issue; the non-linear shift from augmentation to displacement remains unmodeled, leaving any Pigouvian intervention without a clear target.

My View on the Debate

The real issue connecting these papers is a severe institutional blind spot: we are trying to measure a new kind of technological shock using outdated industrial metrics. Our policy frameworks are built to respond to visible economic pain, like factory closures or spiking unemployment claims. But AI displacement is different. It happens at the hiring margin, as entry-level careers that simply never start, and gets absorbed as displaced professionals quietly take lower-paying jobs. Because this occupational downgrading never shows up in official joblessness statistics, the demand trap [Hemenway Falk and Tsoukalas \(2026\)](#) warn about could easily lock in before anyone notices. The most urgent task for researchers isn't just building better theoretical models, but engineering granular, early-warning indicators, like credential mismatch and entry-level hiring ratios, so we can actually track this shadow displacement before it sets in.

Furthermore, this crisis is complicated even further by the political reality of who is actually

at risk. The Penn/BU paper prescribes a sector-neutral automation tax to fix the economic equilibrium, but this completely ignores how high-skill workers behave. When highly educated, politically networked professionals face displacement, they rarely lobby for broad macroeconomic taxes. Instead, they could also use their influence to push for sector-specific protections, like stricter copyright laws on training data or new occupational licensing barriers. The political fallout from AI won't be a unified push for a Pigouvian tax; it will likely be a fragmented push for defensive, guild-like restrictions.

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